

PRESENTATION ON

AI-DRIVEN SPORT ANALYTICS

PLATFORM

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INTRODUCTION

AI-Driven Sports Analytics Platform

Sports analytics has revolutionized how athletes train, compete, and recover. Our platform leverages artificial intelligence and machine learning to provide comprehensive insights for performance optimization and injury prevention.

Key Highlights:

- Real-time performance tracking and analysis
- Predictive analytics for injury risk assessment
- Sport-specific metrics and insights
- Role-based access for different stakeholders

MOTIVATION

- **Automated Insights:** AI turns complex data into quick, accurate decisions.
- **Injury Prevention:** Early risk detection minimizes injuries and recovery time.
- **Personalized Performance:** Customized training improves results and longevity.
- **Competitive Advantage:** Data-driven strategies give teams a measurable edge.



Primary Aim

The primary aim of this project is to design and develop an AI-powered sports analytics platform that enhances athletic performance and minimizes injury risks through systematic data analysis.

Core Focus Areas

- **Performance Optimization:**
To analyze athlete performance metrics and provide personalized training recommendations for improved results.
- **Injury Prevention:**
To identify potential injury risks at an early stage using predictive analysis and support effective workload and recovery management.
- **Data-Driven Insights:**
To generate meaningful insights from sports data using analytical models and visual dashboards for better decision-making.
- **User-Centric Design:**
To develop a secure, role-based, and user-friendly system accessible across web and mobile platforms.

APPLICATIONS

An AI-based sports analytics system that helps players, managers, and data analysts improve performance and reduce injury risks through data-driven insights.

- **Player**
Enables athletes to track personal performance, monitor health metrics, follow personalized training plans, and receive injury-risk alerts.
- **Manager**
Supports team oversight through performance monitoring, workload management, player comparison, and data-driven decision making.
- **Data Analyst**
Provides tools for analyzing performance data, identifying trends, generating reports, and supporting strategic insights.

LITERATURE SURVEY

PAPER 1:

“AI and Machine Learning Applications in Sports Performance Analysis”

Publication and Year	Author name	Approach (Methodology)	Advantages	Limitations
Journal of Engineering Research, 2025 [1]	Mutawa, A. M., Kumar, K. V. R., Murugappan, M.	Video sequence analysis using computer vision; OpenPose-based 3D pose estimation; CNN + ML classifiers	Uses real match data; predicts deceptive movements; improves performance	Offline analysis; single-player study; no real-time live prediction

PAPER 2:

“Wearable Sensor Data Analytics for Athlete Performance and Injury Prevention”

Publication and Year	Author name	Approach (Methodology)	Advantages	Limitations
IEEE Access, 2024	Bilal Hassan, Clare Clough, Yusra Siddiqi, Rao Faizan Ali, Muhammad Asad Arshed	Learning-to-Rank (LTR) models using listwise ranking and artificial neural networks	Position-specific player ranking, Impartial and performance-focused, Over 94% accuracy	Limited training data per position, Assumes strike rate and average always determine best player, Only considers players with prior data for venue/opponent

PAPER 3:

“Predictive Analytics in Sports: Injury Risk Assessment and Performance Optimization”

Publication and Year	Author name	Approach (Methodology)	Advantages	Limitations
Journal of Sports Analytics, 2024	Yash Agrawal, Kundan Kandhway	Supervised ML classification using dynamic and static features; SHAP-based explainable AI	In-play prediction at every over, 85% accuracy, SHAP reveals feature importance shift during match	Only ODI format, Requires ball-by-ball data, Limited to matches with complete innings

PAPER 4:

“Biomechanics of the Bowling Delivery in Cricket: A Critical Analysis”

Publication and Year	Author name	Approach (Methodology)	Advantages	Limitations
IJNRD, 2023	Abhijit Thander	Biomechanical analysis of bowling delivery in cricket; kinematic and kinetic breakdown; phase-wise technique classification	Deep dive into bowling phases; injury risk insights; technique-specific recommendations	Lab-based studies; lacks real-time data; limited individual variation analysis

PAPER 5:

“Data Science in Sports Analytics: A Review of Performance Optimization and Fan Engagement”

Publication and Year	Author name	Approach (Methodology)	Advantages	Limitations
WJARR, 2024	Ogugua C. Obi et al.	Data science in sports analytics; ML for performance optimization and fan engagement; wearable sensor integration	Covers both athlete and fan analytics; personalized training and injury prevention; real-time biomechanical feedback	Generalized across sports; lacks cricket-specific models; ethical/data bias concerns

PAPER 6:

“Methodology and Evaluation in Sports Analytics: Challenges, Approaches, and Lessons Learned”

Publication and Year	Author name	Approach (Methodology)	Advantages	Limitations
Machine Learning, 2024	Jesse Davis, Lotte Bransen, Laurens Devos, Arne Jaspers, Wannes Meert, Pieter Robberechts, Jan Van Haaren, Maaïke Van Roy	Methodology and evaluation in sports analytics; ML-based indicators (e.g., xG); frameworks for player recruitment, match preparation, and health management	Provides structured framework; bridges sports science and ML; practical examples; indicator reliability; interdisciplinary collaboration	Lack of ground truth; noisy/biased data; small sample sizes; evaluation complexity; non-stationary data

PAPER 7:

“An AI-Driven Approach to Real-Time Sports Analytics and Performance Evaluation”

Publication and Year	Author name	Approach (Methodology)	Advantages	Limitations
2024, IEEE ICSES	J. Mohana, Bharthvajan R, M. Murugesan, Habeeb Omotunde, D. Balaji, A. Santhi Mary Antony	AI-driven sports analytics using CNNs, RNNs, and NLP (BERT); real-time data processing via Apache Spark; integration of structured and unstructured data	85% prediction accuracy, 80% faster analysis, real-time insights, handles 10,000 records/hour, customizable	High dependency on cloud infrastructure, complex model training, limited generalization across sports

PAPER 8:

“The Role of Artificial Intelligence in Sports Analytics: A Systematic Review and Meta-Analysis of Performance “

Publication and Year	Author name	Approach (Methodology)	Advantages	Limitations
MDPI Applied Sciences, 2025	Przemysław Pietraszewski, Artur Terbalyan, Robert Roczniok, Adam Maszczyk, Kajetan Ornowski	Systematic review and meta-analysis of 16 studies using CNNs, LSTMs, RL, and computer vision across 13 sports	Pooled accuracy of 87.78%, broad sport coverage, highlights AI's role in tactical decisions and injury prevention	High heterogeneity ($I^2 = 93.75\%$), limited generalizability from lab to real-world, lack of standardization in evaluation

PAPER 9:

“Artificial Intelligence-Powered Sports Analytics: Enhancing Performance Through Data Science”

Publication and Year	Author name	Approach (Methodology)	Advantages	Limitations
IEEE ICCIES, 2025	Saef Wbaid, Azizov Marufkhuja, Md Afzal, Hilola Pulatova	ML-PA-IP framework combining motion capture, biometric monitoring, and predictive analytics for injury prevention and performance	95.6% performance assessment accuracy, 97.6% injury prediction accuracy, real-time feedback and training optimization, 3D visualization and simulation	Requires high-quality biometric data, complex integration of wearable tech, limited validation across sports

PAPER 10:

“Orientation and Decision-Making for Soccer Based on Sports Analytics and AI: A Systematic Review”

Publication and Year	Author name	Approach (Methodology)	Advantages	Limitations
IEEE/CAA Journal of Automatica Sinica, 2024	Zhiqiang Pu, Yi Pan, Shijie Wang, Boyin Liu, Min Chen, Hao Ma, Yixiong Cui	Systematic review of soccer analytics using OODA loop; covers real-world and virtual domains; includes GRF, RL, CV, and NLP	Bridges real and virtual soccer analytics, highlights DRL and simulation platforms, proposes OODA loop for decision modeling	Limited real-world deployment of virtual strategies, sparse reward modeling challenges, interpretability and ethical concerns

PAPER 11:

“AI-Driven Decision Support Systems for Injury Prediction and Performance Monitoring in Sports”

Publication and Year	Author name	Approach (Methodology)	Advantages	Limitations
International Journal of Research and Technology (IJRT), Volume 13, Issue 1, February 2025	Dr. Sumit Jain	Decision Support System, Multimodal data collection using wearable sensors, video analytics, and medical records	High prediction accuracy (up to 93.8%), Early injury detection and prevention, Real-time player performance monitoring, Personalized training and rehabilitation plans	High data requirements, implementation cost, computational complexity, and player data privacy concerns.

PAPER 12:

“Artificial Intelligence Applications in Sports Biomechanics: A Scoping Review of Wearables and Motion Analysis”

Publication and Year	Author name	Approach (Methodology)	Advantages	Limitations
Bioengineering (MDPI), 2024.	Marouen Souaifi, Wissem Dhahbi, Nidhal Jebabli, Halil İbrahim Ceylan, Manar Boujabli, Raul Ioan Muntean, and Ismail Dergaa.	A PRISMA-ScR–based scoping review examining artificial intelligence applications in sports biomechanics, with emphasis on wearable technologies and motion analysis for performance and injury assessment.	High-precision motion analysis with real-time feedback for technique improvement, enhanced injury risk prediction, and flexible in-field monitoring enabled by wearable technology.	High cost, advanced AI systems, wearable sensors, small sample sizes, limited generalizability, technical complexity, specialized expertise required.

PAPER 13:

“The Role of Artificial Intelligence in Sports Performance Analysis, Injury Prevention, and Talent Identification: A Systematic Review”

Publication and Year	Author name	Approach (Methodology)	Advantages	Limitations
Geopolitical, Social Security and Freedom Journal, Volume 7, Issue 1, 2024	Andrea Pisaniello, University of Cassino and Southern Lazio	Systematic review of peer-reviewed studies (2010–2023) from PubMed, IEEE Xplore, Google Scholar, and Scopus, focusing on AI-based performance analysis, injury prevention, and talent identification.	Injury prevention through biomechanical pattern analysis, AI-driven talent identification, and objective player evaluation beyond traditional scouting	Interpretability issues (“black-box” models), ethical and privacy risks in biometric data collection, and algorithmic bias affecting fair talent evaluation.

PAPER 14:

“Real-Time Match-Fixing Detection in Football Using Multimodal Machine Learning Models”

Publication and Year	Author name	Approach (Methodology)	Advantages	Limitations
Scientific Reports (Nature Portfolio), 2024.	Changgyun Kim, Jae-Hyeon Park, and Ji-Yong Lee.	Real-time match-fixing detection system using football league betting odds data, employing a multimodal AI ensemble (SVM, Random Forest, Logistic Regression, and KNN) with time-series analysis.	High Accuracy, Real-Time Monitoring, Nuanced Classification, Robust Data Integration	Data Imbalance, Complexity of Betting Markets, Static Data Exclusion, Narrow Geographic Scope for Training, Narrow Geographic Scope for Training

PAPER 15:

“Real-Time Possessing Relationship Detection for Sports”

Publication and Year	Author name	Approach (Methodology)	Advantages	Limitations
Chinese Control Conference, 2020	Yinda Xu, Yonggang Peng	Algorithm framework combining visual object tracking (SiamFC++), human pose estimation (CenterNet), IoU tracking, and heuristic relationship reasoning for detecting possession in sports videos	Real-time performance (20+ FPS), effective possession detection (hand/foot), robust tracking with diverse training data, validated on public video datasets	Pose continuity issues, simple IoU tracker limits robustness, reasoning restricted to two relationships, challenges with fast motion and occlusion

PAPER 16:

“Scalable Infrastructure for Efficient Real-Time Sports Analytics”

Publication and Year	Author name	Approach (Methodology)	Advantages	Limitations
ACM ICMI Companion, 2020	Håvard D. Johansen, Dag Johansen, Tomas Kupka, Michael A. Riegler, Pål Halvorsen	PMSys smartphone-based athlete monitoring system; cloud microservices; AI prediction using LSTM for readiness and wellness data	Real-time athlete monitoring, scalable cloud infrastructure, privacy-preserving data collection, promising AI predictions, widely adopted in elite teams	Limited dataset size, subjective self-report variability, high development/maintenance overhead for apps, dependency on cloud services

PAPER 17:

“Analysis of Cricket Ball Type and Innings on State Level Cricket Batter's Performance”

Publication and Year	Author name	Approach (Methodology)	Advantages	Limitations
Frontiers in Psychology, 2019	Jonathan Douglas Connor, Wade H. Sinclair, Anthony S. Leicht, Kenji Doma	Statistical analysis of batting performance across 60 Sheffield Shield games; comparison of Kookaburra™ vs. Duke™ balls; first vs. second innings performance	First innings showed higher runs, balls faced, boundaries, and centuries; ball type had no significant effect; pace bowlers more effective with Duke™, spin bowlers with Kookaburra™	Decline in batting performance during second innings, order effect in ball usage across seasons, limited mechanistic data on pitch deterioration, small sample constraints

PAPER 18:

“Data Analytics in the Game of Cricket: A Novel Paradigm”

Publication and Year	Author name	Approach (Methodology)	Advantages	Limitations
Procedia Computer Science, 2022	Varad Vishwarupe, Mangesh Bedekar, Prachi M. Joshi, Milind Pande, Vishal Pawar, Prathamesh Shingote	IoT-based cricket bat sensors (Stance Beam) used to capture bat speed, angles, timing index; case study with real-life net session; categorization of shots into timing index classes (Early, Late, Timed, Missed, Well-Left)	Real-time shot analysis, timing index integration, IoT-enabled data capture, actionable training insights, improved batting mechanics understanding	Small sample size, short session duration, device normalization issues, follow-through metric less reliable, preliminary validation required

PAPER 19:

“Applications of Machine Learning in Cricket: A Systematic Review”

Publication and Year	Author name	Approach (Methodology)	Advantages	Limitations
Machine Learning with Applications, 2022	Indika Wickramasinghe	Systematic review of 59 publications (2001–2021) applying ML in cricket; PRISMA protocol; analysis of research areas including outcome prediction, player classification, batting/bowling style, umpire decisions, pitch behavior, and score prediction	Overview of ML in cricket, key research areas identified, growth since 2010 noted, techniques like Random Forest, Decision Trees, kNN, Naïve Bayes, and regression applied, future directions outlined	Data heterogeneity across formats, limited datasets in some studies, inconsistent feature extraction, variable accuracy reporting, lack of standardized evaluation frameworks

PAPER 20:

“Deep Learning in Cricket: A Comprehensive Survey of Shot Detection and Performance Analysis”

Publication and Year	Author name	Approach (Methodology)	Advantages	Limitations
IRJAEH, 2024	Milind Mahajan, Shreekar Kulkarni, Mitesh Kulkarni, Abhishek Sabale, Aakanksha Thaka	Systematic survey of 10 papers on cricket shot detection and performance analysis; methods include CNNs, LSTMs, transfer learning, multimodal fusion (video, audio, image); evaluation with accuracy, precision, recall, F1-score	Review of deep learning in cricket, CNN/LSTM effectiveness, multimodal integration, high shot detection accuracy, potential for automatic commentary and performance evaluation	Dataset scarcity, limited generalization across formats, reliance on handcrafted preprocessing, varying input modalities, need for real-time implementation and broader validation

PAPER 20:

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IRJAEH, 2024	Milind Mahajan, Shreekar Kulkarni, Mitesh Kulkarni, Abhishek Sabale, Aakanksha Thaka	Systematic survey of 10 papers on cricket shot detection and performance analysis; methods include CNNs, LSTMs, transfer learning, multimodal fusion (video, audio, image); evaluation with accuracy, precision, recall, F1-score	Review of deep learning in cricket, CNN/LSTM effectiveness, multimodal integration, high shot detection accuracy, potential for automatic commentary and performance evaluation	Dataset scarcity, limited generalization across formats, reliance on handcrafted preprocessing, varying input modalities, need for real-time implementation and broader validation

PAPER 21:

“Artificial intelligence in sport: A narrative review of applications, challenges and future trends”

Publication and Year	Author name	Approach (Methodology)	Advantages	Limitations
<i>Artificial intelligence in sport: A narrative review of applications, challenges and future trends</i> (2025, Journal of Sports Sciences)	Diwei Zhou, Justin W.L. Keogh, Yingliang Ma, Raymond K.Y. Tong, Abdul R. Khan, Nicholas R. Jennings	Narrative review synthesizing AI applications in biomechanics, performance enhancement, sports medicine, health monitoring, coaching, and talent identification. Discusses machine learning, deep learning, NLP, computer vision, AR/VR, ensemble learning.	Broad coverage across multiple domains, Practical tools highlighted (wearables, apps), Inclusive focus (Para athletes), Future trends identified (digital twins, XAI)	Ethical/privacy concerns, High cost and complexity of adoption, Limited accessibility outside elite sport, “Black box” trust issues, Accountability challenges

RESEARCH GAP

- Lack of real-time AI-driven analytics integrating performance + injury prediction.
- Limited role-based platforms combining athlete, manager, and analyst insights.
- Insufficient use of predictive ML models for proactive injury prevention.

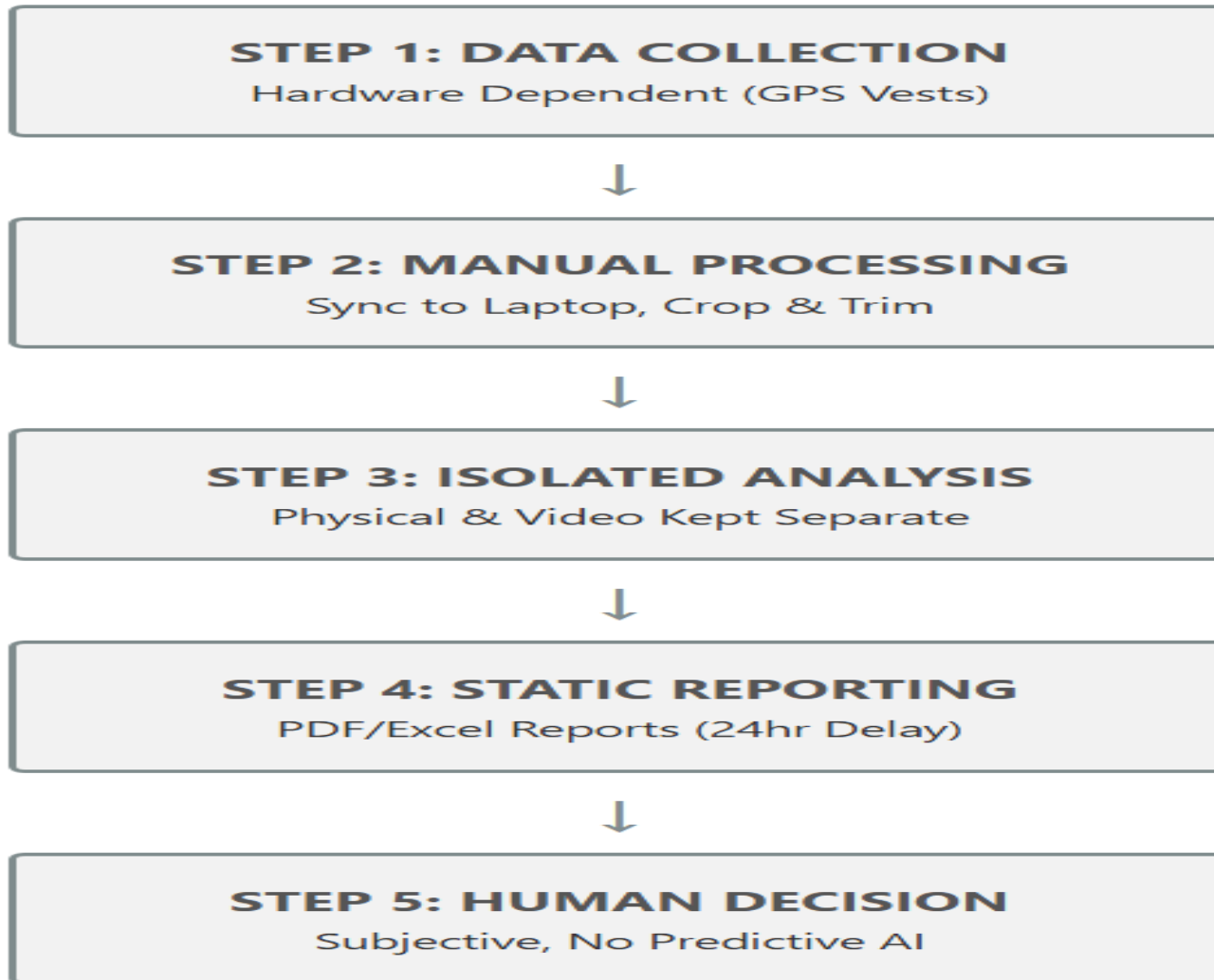
OBJECTIVES

- Develop an AI-powered sports analytics platform for performance optimization.
- Implement injury risk prediction using ML and historical data.
- Provide role-based dashboards with actionable data insights.

METHODOLOGY

- Holistic Data Integration: We combine physical, tactical, psychological, environmental, and historical data to get a complete and accurate performance analysis.
- Multi-Dimensional Analysis: Performance and injury risk are evaluated using multiple weighted factors instead of relying on a single metric.
- Ensemble & Adaptive AI: Multiple AI models work together to improve prediction accuracy beyond traditional single-model systems.
- Continuous Learning & Explainability: The system learns from real outcomes and clearly explains predictions to ensure trust and better decision-making.

EXISTING WORK FLOW



PROPOSED WORK FLOW



IMPLEMENTATION

- Frontend: React.js, Tailwind CSS, Chart.js dashboards.
- Backend: Node.js, Express.js, REST APIs, JWT authentication.
- AI Module: Python-based ML models for prediction and recommendations.

CONCLUSION

- Platform enables data-driven decision-making in sports.
- Improves athlete performance and injury prevention.
- Scalable solution for multiple sports and stakeholders.

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THANK YOU...